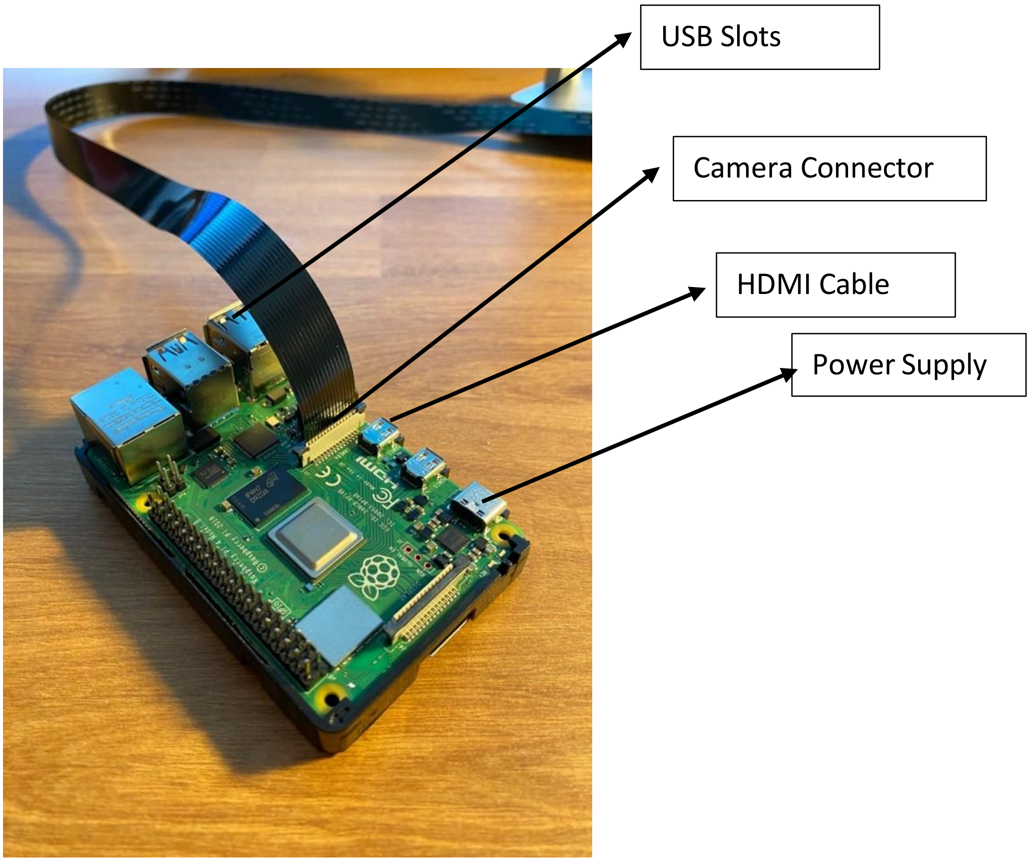
ISPEED Microscopy Kit Setup

Important Notes

1. When plugging in/plugging out the camera, please ensure that the Raspberry Pi is powered off.
2. When handling sensitive components such as lenses and cameras, ensure that gloves are worn, and the components are placed gently on the table.

Setting up Raspberry Pi

1. Plug in the HDMI cable, SD card, power cable, USB mouse and USB keyboard into the Raspberry pi.

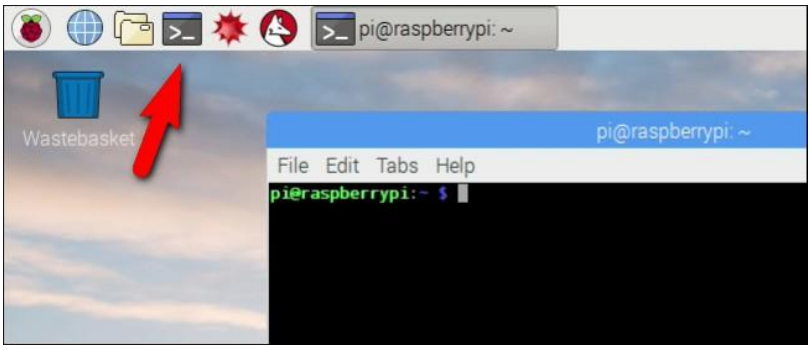


2. The SD card should already contain Raspberry Pi OS. If not, install Raspberry Pi OS using Raspberry Pi Imager on another computer.

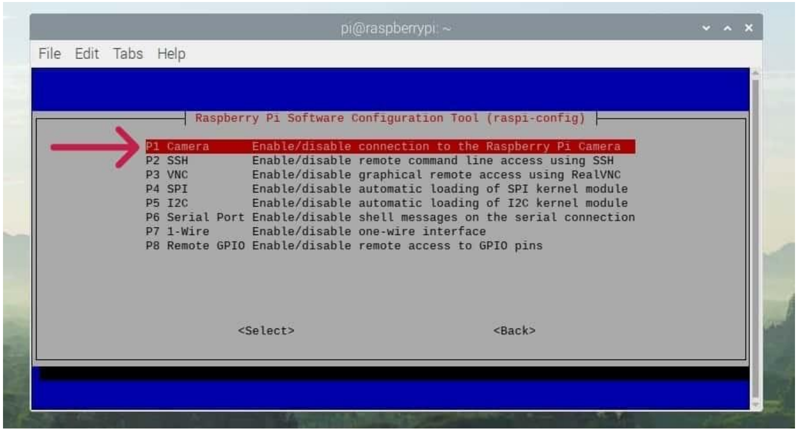
3. Once the power cable is plugged in, the Raspberry Pi should power on. Wait until the desktop appears.

4. Once you see the desktop, open the terminal (black icon).

5. On newer Raspberry Pi OS versions, the camera interface may already be enabled and the "Camera" option may not appear in raspi-config. If needed, type the following command and press Enter: `sudo raspi-config`



6. If the Camera option is available, go to “Interface Options” > “Camera.”



7. Click enter and then enable camera.
8. Exit Raspi-config and reboot the Raspberry pi: `sudo reboot`

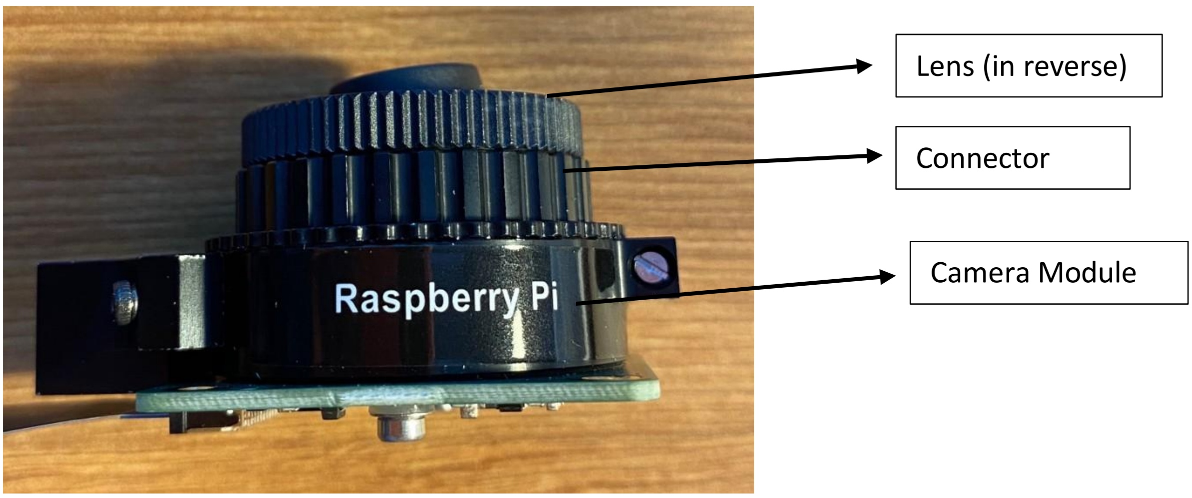
Building the Microscope Lens

1. *Make sure the raspberry pi board is powered off before completing this section.*

2. Please wear gloves for this process. Please be careful to not touch the lens and the camera sensor directly. Always grip the sides.

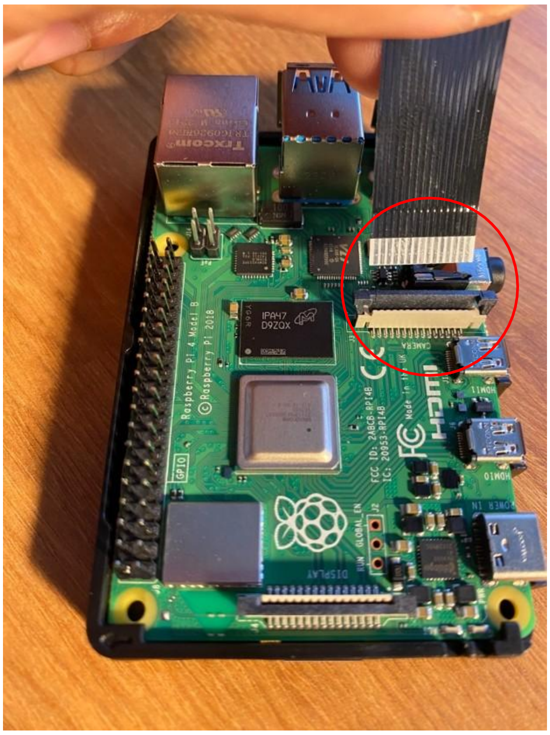
3. Remove the cap from the macro lens and screw on the lens holder.

4. Then screw on this system onto the camera.



5. Connect the ribbon cable to the camera by lifting the light brown tab, slotting the cable in such that the metal contacts are aligned (please do not connect them in the wrong way).

6. Push the tabs down to lock it in place. Please be gentle and do not yank or force the tabs in.

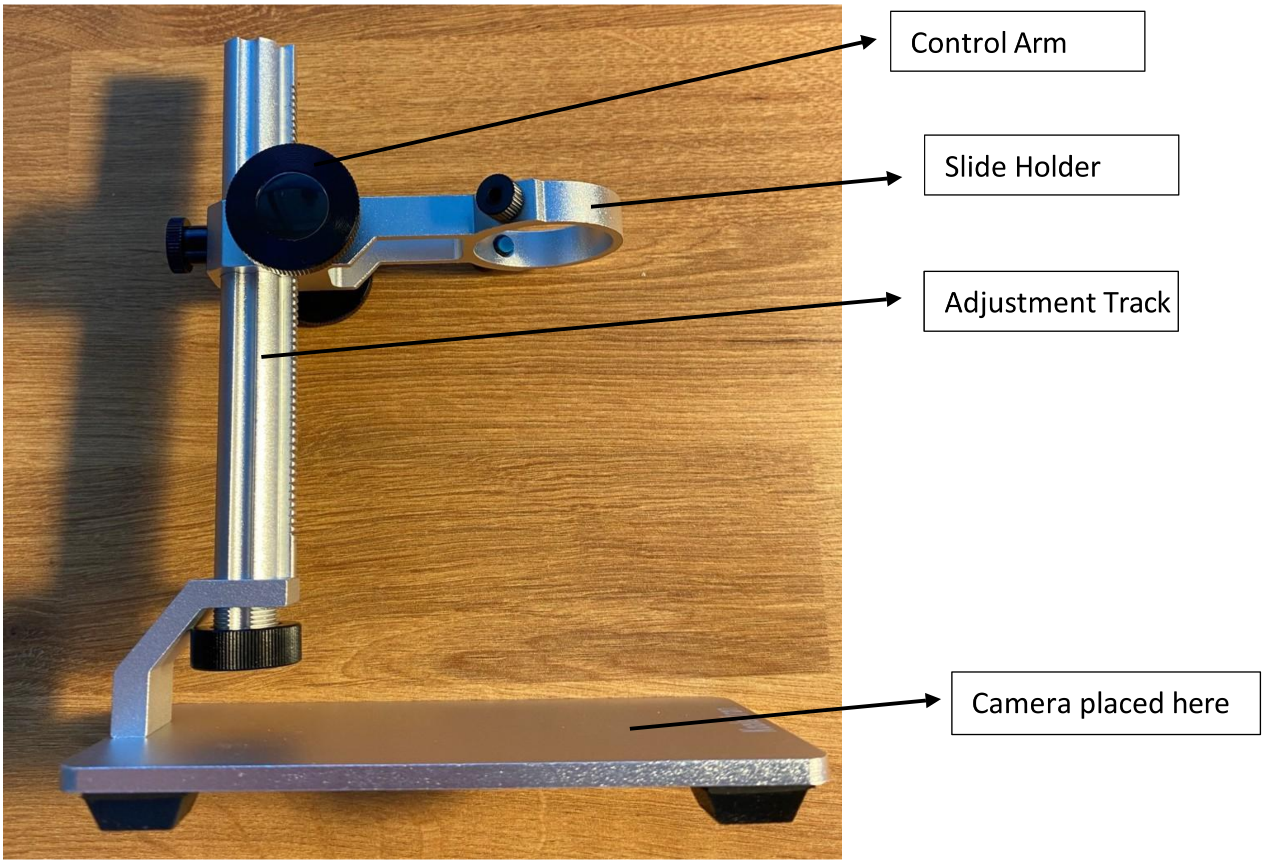


7. Connect the other end of the ribbon cable
to the connector labelled “camera” (please don’t insert it into the connector labelled “display”, they look very similar).

8. Power on the raspberry pi.

Setting up the microscope stage

1. Assemble the microscope stand according to the provided instructions. Ensure that the arm moves up and down smoothly when the black knob is turned. If it does not, double-check that the pieces are installed in the correct orientation.



2. Place a box or support platform on the base of the stage. Then place the camera above it so that the stage can be moved down toward the camera.
   Note: The required box height may need to be adjusted to obtain a focused image.Place a slide on top of the circular holder.

3. Place a slide on top of the circular slide holder.

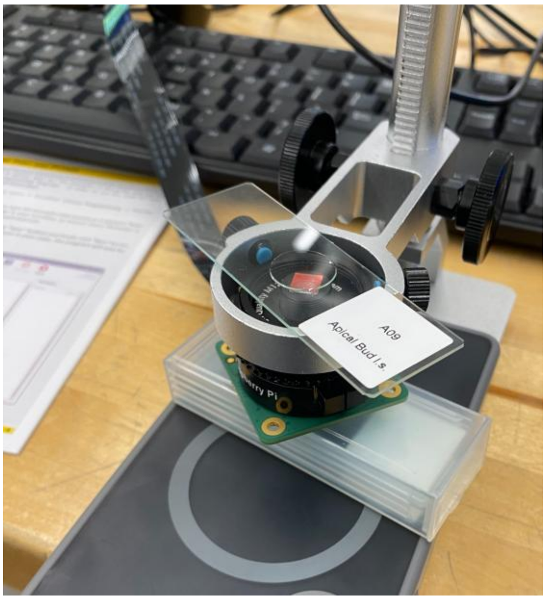
4. Turn the black knob to lower the stage toward the camera until the sample comes into focus.

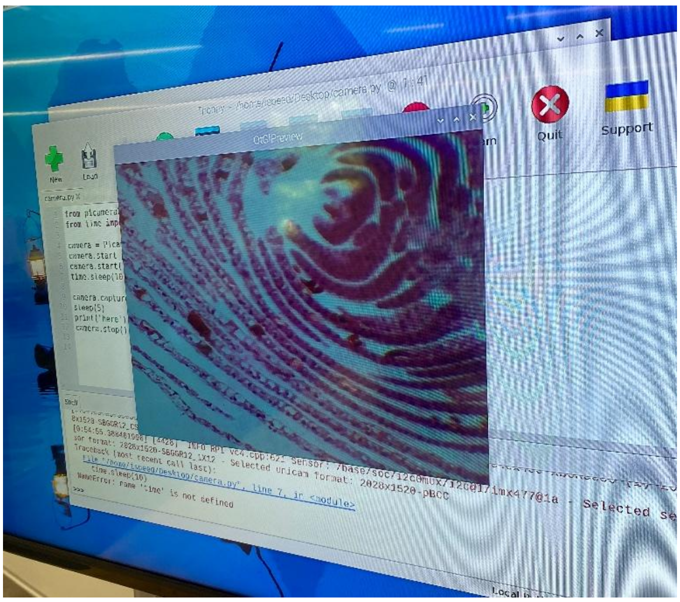
5. To view a live camera preview while focusing, open the terminal on the Raspberry Pi.

6. Type the following command and press Enter: `rpicam-hello -t 0`
   A live preview will appear and help you adjust the camera-to-slide distance until the image is focused.

7. To stop the live preview, press Ctrl+C.

Completed Microscope Setup





Acquiring images via Python Code

1. Open Thonny Python IDE (should already be pre-installed)
2. Open a new file save it as **camera.py** (important to **never** save the file name as picamera.py)
3. Enter the following code:

```
from picamera2 import Picamera2, Preview
from time import sleep

camera = Picamera2()
camera.start_preview(Preview.QTGL)
camera.start()
sleep(5)
camera.stop_preview()
camera.stop()
camera.close()
```

4. Once camera is working, amend your code as follows:

```
from picamera2 import Picamera2, Preview
from time import sleep

camera = Picamera2()

try:
    camera.start_preview(Preview.QTGL)
    camera.start()
    sleep(5)
    camera.capture_file("/home/pi/Desktop/image.jpg")

finally:
    camera.stop_preview()
    camera.stop()
    camera.close()
```

   Note: It is important to let the camera sleep for at least 2 seconds before capturing an image so that the camera sensor has time to sense the light levels.

5. Image resolution and adding text to your image can be done using the following guide: https://projects.raspberrypi.org/en/projects/getting-started-with-picamera/7

Support and Additional Resources

All the instructions, code and parts lists used are available for free at this link:

https://github.com/ShamsNafisaAli/ISpeed-Microscopy